Set Vision Policy: Learned Perception to Browser Action

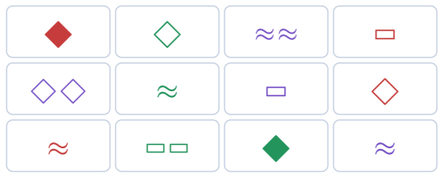
LOCAL VISION-TO-ACTION POLICY

41,918 PARAMS | NO ATTENTION

Compact CNN - four specialists - differentiable relation scorer - local WASM deployment

1 INPUT OBSERVATION

Live board screenshot



Screenshot	H x W x 4 RGBA
Cards	12 clickable rectangles

2 DATA PREPROCESSING PIPELINE

Capture — visible browser viewport

→ **Crop** — 12 DOM rectangles, geometry only

→ **Area average** — integer source bins, anti-alias stable

→ **Normalize** — RGB / 255, clamp 0..1

→ **Pack** — 12 x 1,890, f32 -> WASM

SOURCE CROP	CARD TENSOR	BOARD TENSOR	CONTRACT
variable W x H x 4	[3, 21, 30]	[12, 3, 21, 30]	browser = trainer fixture

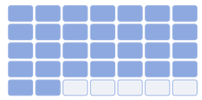
PROCEDURAL TRAIN SOURCE

Production Card renderer | four themes | multiple raster widths | seeded color, texture, geometry, and lighting variation

3 INTERMEDIATE REPRESENTATIONS



RGB card



shared feature map

Input	[N, 3, 21, 30]
Shared	[N, 12, 11, 15]
Per head	[N, 8, 6, 8]
Embedding	[N, 24]
Attr logits	[N, 4, 3]

BOARD CACHE

12 cards encoded once
144 attribute logits

6 TRAINING OBJECTIVES

SET LOSS

CE([0, score], y-set)

AUXILIARY LOSS

4 x mean CE(attribute logits, labels)

$$\mathcal{L} = \mathcal{L}_{\text{set}} + 4\mathcal{L}_{\text{attr}}$$

Optimizer	AdamW | lr 8e-4 cosine | wd 1e-4
Budget	20k x 64 = 1.28M triples

8 ROBUSTNESS & ANALYSIS

Procedural split protocol

TRAIN
272 atlases
seeds 0-16
widths 128, 160, 224, 280

RESEARCH V5
240 atlases
seeds 11001-11002
6 capture profiles + DPR 2

SEALED V5
240 atlases
seeds 12001-12002
unseen widths and profile mix

All splits span four themes. Research and final seeds and widths are absent from training. The sealed holdout is evaluated once, then retired.

DIAGNOSTICS + GUARD

Relation score/margin and attribute entropy/margin feed a research-fitted abstention gate. Repeated frames and post-click pixel change are also required.

~~Finite envelope claim - not proof of universal 100% reliability.~~

4 CORE ARCHITECTURE

CARD BATCH — [N, 3, 21, 30], RGB f32

→ **SHARED CONV** — 5 x 5 | stride 2 | pad 2, 3 -> 12 channels | 912 params

→ **SHARED ACTIVATION** — exact-erf GELU, [N, 12, 11, 15]

↓ fan-out: same shared map enters four independent learned branches ↓

COUNT	COLOR	SHAPE	FILL
Conv2d: 12 -> 8 | k3 s2 p1, [N, 8, 6, 8]	Conv2d: 12 -> 8 | k3 s2 p1, [N, 8, 6, 8]	Conv2d: 12 -> 8 | k3 s2 p1, [N, 8, 6, 8]	Conv2d: 12 -> 8 | k3 s2 p1, [N, 8, 6, 8]
GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params	GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params	GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params	GELU ↓ PROJECTION: flatten 384 -> 24 -> 3, GELU + softmax | 10,187 params

ENUMERATE — 12 choose 3 = 220 candidate actions

→ **TRIPLE EVIDENCE** — for each attribute: p(i) + p(j) + p(k), shape [220, 4, 3]

→ **ATTRIBUTE MLP** — shared 3 -> 32 -> 1, GELU \| x4, 161 params

→ **SET MLP** — concat 4 -> 16 -> 1, GELU, 97 params

→ **ACTION** — argmax score -> [i, j, k]

5 ATTRIBUTE SPECIALIST OUTPUTS

COUNT	COLOR	SHAPE	FILL
1 | 2 | 3 p = [p1, p2, p3]	red | green | purple p = [pr, pg, pp]	diamond | squiggle | oval 3 learned logits	solid | striped | outline 3 learned logits

Auxiliary labels make each branch inspectable. The relation scorer consumes probabilities, not argmax labels, so uncertainty and gradients remain continuous.

7 INFERENCE & DEPLOYMENT

BROWSER — capture + crop, area resample

→ **WASM** — FrankenTorch, embedded .ftsv

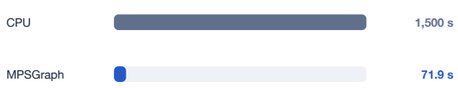
→ **GUARDED ACTION** — two stable reads, click + verify change

Checkpoint	330 KB .ftsv
Generated WASM	2.5 MiB
Cached inference	2.502 ms / board
With pixel transfer	2.767 ms / board

9 BENCHMARKS & EVALUATION

487 / 4,096	1,085 / 10,000	99.84% @ 24.62%	4 / 12
v5 research raw failures	v5 sealed raw failures	sealed guarded accuracy / coverage	guarded browser actions verified

20k-step training time - about 20.9x faster



Board inference latency



10 AUTORESEARCH LOOP & PROMOTION GATES

HYPOTHESIS — one falsifiable change

→ **TRAIN** — fixed 20k-step budget

→ **RESEARCH** — 4,096 boards, keep / discard

→ **SEALED** — 10,000 boards, exactly once

→ **PROMOTE** — same checkpoint -> WASM

Selection order

1	fewer board failures
2	higher triple accuracy
3	lower latency, then size

Reproducibility controls

Selected seeds	490009 / 0x369dea0f31a53f85
Batch	64 balanced triples
Replications	8 fixed seed pairs

11 MODEL FACTS

Policy	compact CNN
Attention	none
Memory	none
Runtime solver	none
Output	3 card indices
Parameters	41,918
Actions	220 / board

LEGEND - MODULE COLORS

shared CNN · relation scorer · runtime / state · specialists · loss / training · browser action

TENSOR / SHAPE GUIDE

N	card count or batch cards
C, H, W	channels, height, width
[N, 4, 3]	four attributes x three values
[220, 4, 3]	all triple evidence vectors

GRADIENT / DATA FLOW

→ solid: inference data flow ↓ branch / supervision direction

Training loss backpropagates through both relation MLPs, probability sums, each specialist softmax and classifier, specialist convolutions, and the shared convolution. There is no nondifferentiable gate between loss and specialists.

IMPLEMENTATION SOURCES

Policy and browser path: `vision-policy/src/lib.rs`
Fast training graph: `vision-mpsgraph/src/policy_graph.rs`
Pixel contract: `extension/preprocess.js`
Protocol and results: `vision/AUTORESEARCH.md`, `vision/RESEARCH_LOG.md`

Diagram generated from editable Typst source. Metrics measured on Apple M5 Pro, 2026-08-18.